SUPPLY CHAIN ANALYSIS REPORT

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1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfilment across multiple regions. The analysis covers 172,765 orders spanning January 2015 through January 2018, focusing on identifying root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for improvement.

The headline finding is stark: 54.71% of all orders are delivered late, costing the business $2.1 M in eroded profitability. While total profit across fulfilled orders reached $7.5M, in the persistent late-delivery problem represents both a significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they became late.

Key Finding	Metric
Overall Late Delivery Rate	**54.71%**
On-Time Delivery Rate	**45.29%**
Total Orders Analysed	**172,765**
Total Profit (profitable orders)	**$7.5M**
Profit at Risk (delayed orders)	**$2.1M**
Predictive Model Accuracy (RF)	**74%**

2. Business Context and Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability and an inability to make reliable commitments to buyers at the point of purchase.

Analytical Objectives:

- Understanding the current state of delivery performance across all dimensions (region, mode, time, segment)
- Quantify the financial impact of delays on order profitability
- Identify the primary operational bottlenecks driving late deliveries
- Build a predictive model to flag high-risk orders before they are shipped
- Deliver actionable recommendations to improve on-time delivery and profitability.

3. Key Performance Indicators (KPIs)

The following KPIs were compiled directly from the cleaned dataset to establish a performance baseline.

172,765 Total Orders	**94,523** Late Deliveries	**54,71%** Late Delivery %	**42.59%** On Time Delivery %
$7.5M Total Profit (Profitable Orders)	**$2.1M** Profit Risk (Delayed Orders)	**3 Days** 90th Percentile Delay	**$22.03** Average Order Profit

Late Delivery Risk at 54.71%	More than half of all orders arrive late. This is not an edge-case problem. It is the default experience for the majority of the customers.

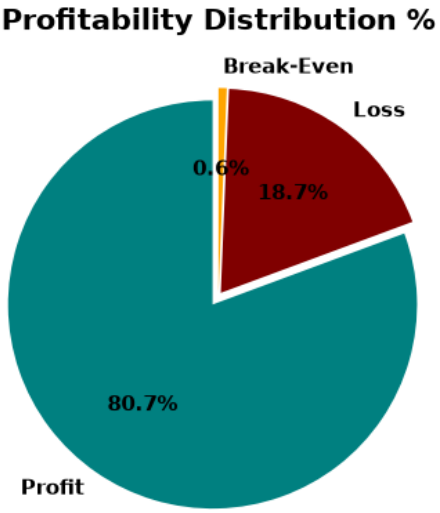
$2.1M Profit at Risk	Orders that experienced delays collectively generated $2.1M in profit that is under constant pressure from further operational deterioration.

90th Percentiles Delay = 3 days	Even the most extreme cases are constrained within 3 days of delay, suggesting the problem is systematic and process-driven rather than catastrophic.

4. Profitability Analysis

4.1 Profitability Distribution

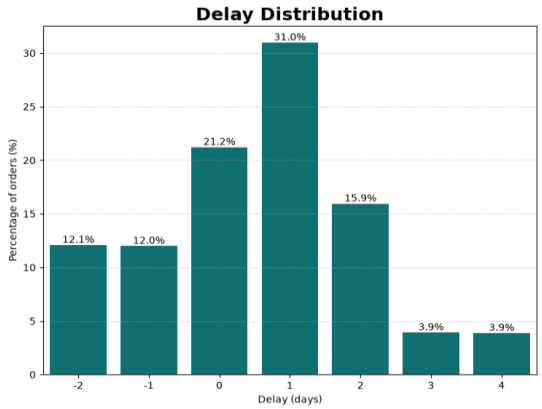
Order level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.

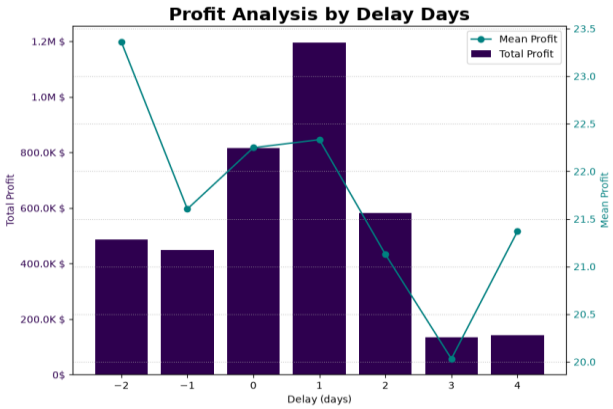


Tier	Definition	Share
Profit	Order Profit per Order > 0	80.7%
Loss	Order Profit per Order < 0	18.7%

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31% of all orders arrive exactly 1 day late — the single largest cohort. Combined, orders delayed by 1-4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.

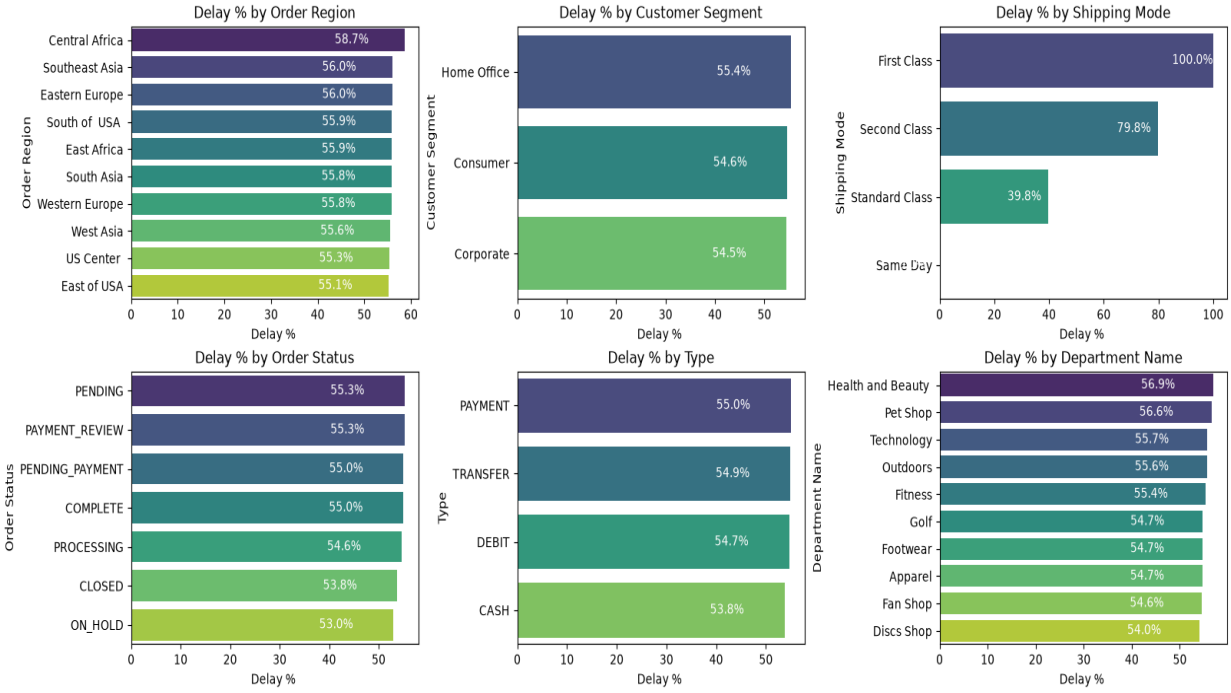




A critical observation: mean profit per order remains stable at approximately $21-$23 across all delay levels. The profitability problem is therefore driven by the volume of delayed orders, not by individual orders economics making systemic throughput improvement the highest-level intervention.

5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfilment process breaks down most severely.



	First Class: 100% delay rate. Second class: 79.8%. Standard Class: 39.8%. Same

Shipping Mode is the #1 Lever	Day:0%. The mode assignment logic is the most impactful single variable to fix.

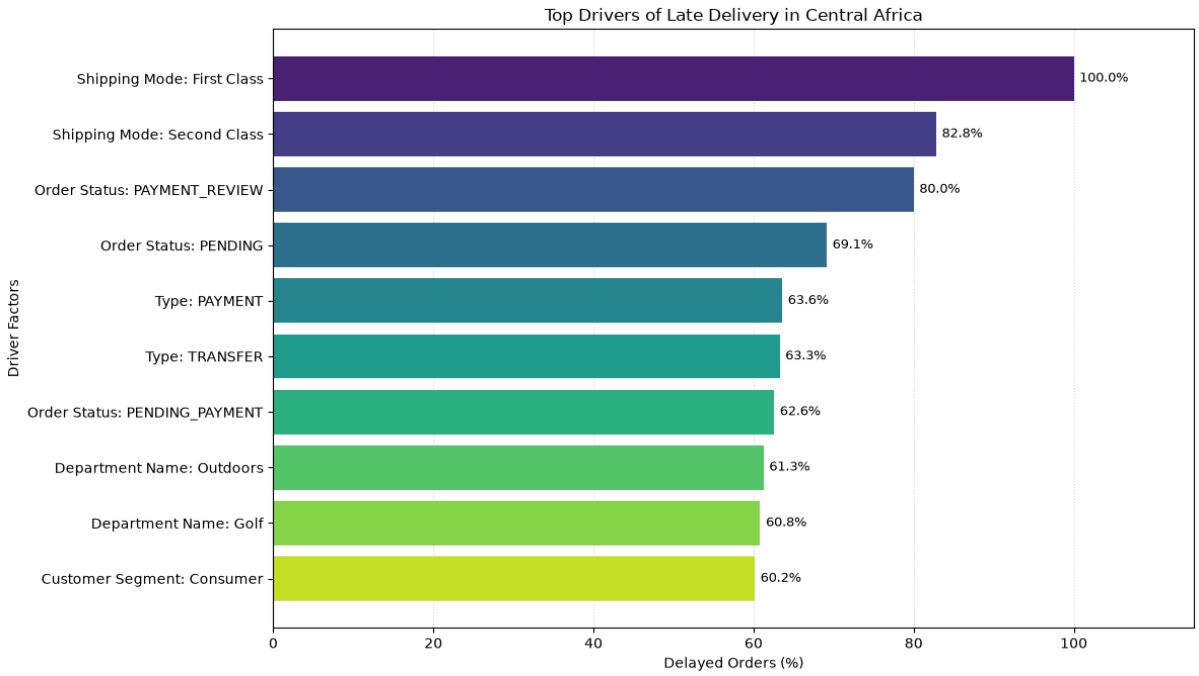
Regional Spread is Narrow (55-59%)	Central Africa leads at 58.7% but all regions cluster between 55-59%. This rules out a localised logistics failure and confirms a company-wide systemic issue.

Customer Segments Nearly Identical (54.5-55.4%)	No segment receives preferential service. All customers — Consumer, Corporate, Home Office — experience the same broken delivery promise.

Health and Beauty (56.9%) and Pet Shop (56.6%) lead by Department	These departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these product categories.

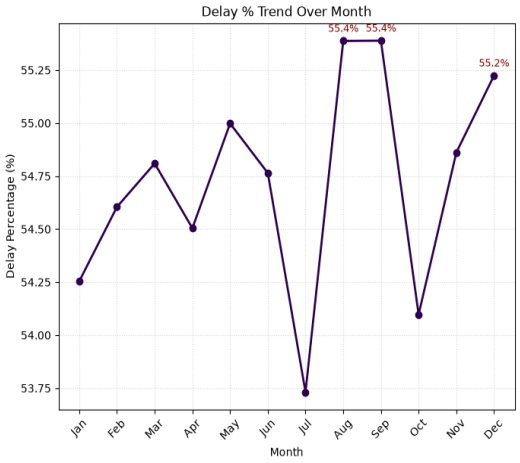
6. Root Cause Analysis

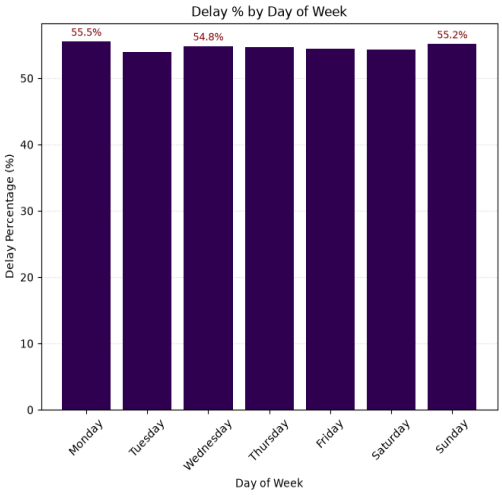
Central Africa was selected for deep-dive root cause analysis as the highest delay region (58.7%). The chart below ranks the top 10 driver factors by delay percentage within this region.

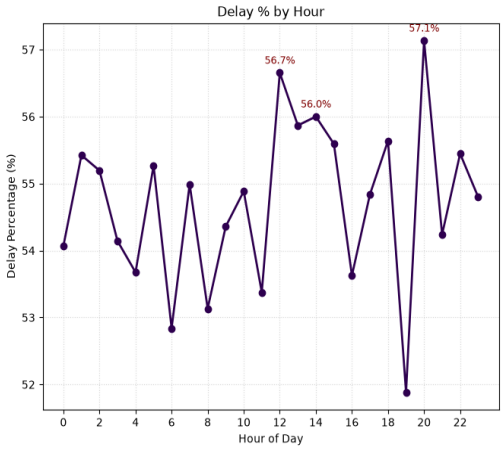


7. Time-Based Delay Patterns

Three temporal dimensions were analysed: month of year, day of week and hour of day. While delay rates are relatively stable across all dimensions (53-57%), specific peaks highlight opportunities for targeted capacity planning.







Peak Delay Months: August and September (55.4%), December (55.2%)	August and September are the highest months. With December close behind. These reflect mid-year promotions and Q4 Holiday surge overwhelming fulfilment capacity. July represents notable low-point (-53.75%), confirming seasonal variation is plannable.

Day-of-Week Variance is minimal	Monday (55.5%) and Sunday (56.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day-of-week is not a meaningful lever for operational intervention.

Intra- Day Peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)	Late evening orders (Hour 21) have the highest delay rate 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak-volume hours.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_Delivery_risk = 1). The pipeline included frequency encoding of categorical variable, stratified train/test split. SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

SPLIT	CLASS 0 (On Time)	CLASS 1 (Late)
Before SMOTE (Train)	59,030	79,182
After SMOTE (Train)	79,182	79,182

Random Forest Classifier-Test Set Performance (34,553 records)

METRIC	CLASS 0 (On Time)	CLASS 1 (Late)	OVERALL (Weighted)
Precision	0.68	0.79	0.74
Recall	0.73	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy	-	-	0.74

74% Overall Accuracy	The model correctly classifies nearly 3 in 4 orders — a strong foundation for production deployment as an order-level late delivery alert system.

Higher Precision on Late Orders (0.79)	When the model predicts an order will be late, it is correct 79% of the time— sufficient to trigger targeted interventions without generating excessive false alarms.

Recall for Late Orders (0.75)	The model captures 75% of all late orders. The 25% miss rate is acceptable for a first-generation alert system, with clear improvement potential.

Class 0 Harder to Predict (0.68)	On-Time deliveries share many characteristics with the late ones given the systemic nature of delays, making them genuinely difficult to distinguish.

9. Strategic Recommendations

Based on the findings across all the analytical dimensions, the following prioritised actions are recommended:

CRITICAL	**Immediately Audit First Class & Second Class Shipping Capacity:** First Class shipping has a 100% delay rate and Second Class 79.8%. These modes operate in complete contradiction to their brand promise. Conduct an immediate carrier SLA audit. Until resolved, consider suspending First Class or repricing to reflect actual performance.
HIGH	**Deploy the Predictive Alert System:** The Random Forest model is ready for production piloting, integrate into the order management system to flag high-risk orders at confirmation and trigger proactive customer communication and priority handling queues.
HIGH	**Resolve Payment Processing Bottlenecks:** Orders in PAYMENT_REVIEW (55.3% overall, 80% in Central Africa) and PENDING_PAYMENT (55%) have significantly elevated delay rates. Introduce automated escalation for stalled payment reviews exceeding 2 hours.
MEDIUM	**Develop Seasonal Surge Capacity Plans:** August, September (55.4%) and December (55.2%) are peak delay months. Build pre-season capacity plans including temporary logistics partnerships, inventory pre-positing and adjusted delivery promise windows.
MEDIUM	**Default to Standard Class for Eligible Orders:** Standard Class achieves a 39.8% delay rate—far superior to First Class and Second Class. Re-evaluate shipping mode assignment logic to default eligible orders to Standard Class, particularly in regions and departments with poor premium mode performance.

MEDIUM	**Investigate High-Delay Departments in Africa:** Outdoors (61.3%) and Gulf (60.8%) in Central Africa show significantly elevated delay rates. Conduct a warehouse-level audit of stock availability, pick-up-and-pack times, and carrier assignment for these product categories.
LOW	**Reduce Loss-Makin Order Share (18.7%):** The loss-making order share warrants a separate pricing and discount review. Combine with delay analysis to identify whether discounted orders are disproportionately late, creating a compounded margin problem.
LOW	**Continuously Retrain the Predictive Model:** Retrain the model quarterly. Explore adding carrier-level data, weather events, and warehouse utilisation features. Target a recall improvement to above 80% on late orders within 6 months.